함수형 Component

Hook

권기웅

1. Hook

1.1 Hook

React 측은 클래스형 Component 보다 함수형 Component 를 사용하기를 권장 :
https://ko.reactjs.org/docs/hooks-intro.html#motivation

1.2 State Hook

1.2.1 useState

Kekeke.js

```
import React, { useState } from "react";

const Kekeke = () => {
  const [ke, setKe] = useState("ㅋ");
  const [cnt, setCnt] = useState(2);
  const laugh = () => {
    setCnt(cnt + 1);
    let t = "";
    for (let index = 0; index < cnt; index++) {
      t += "ㅋ";
    }
    setKe(t);
  }; // nfn

  return (
    <div>
      <h1>{ke}</h1>
      <button onClick={laugh}>웃어</button>
    </div>
  );
};

export default Kekeke;
```

App.js

```
import React from "react";
import Kekeke from "./Kekeke";

const App = () => {
  return <Kekeke />;
};

export default App;
```

1.2.2 useReducer

laugh.js

```
import React, { useReducer } from "react";

function doLaugh(stateee, actionnn) {
  if (actionnn.a === "ㅅㅈ") {
    return "";
  }
  return stateee + actionnn.l;
}
const Laugh = () => {
  const [laugh, setLaugh] = useReducer(doLaugh, "ㅋ");

  return (
    <div>
      <button onClick={() => setLaugh({ a: "ㅊㄱ", l: "ㅋ" })}>ㅋ</button> 
      <button onClick={() => setLaugh({ a: "ㅊㄱ", l: "ㅎ" })}>ㅎ</button> 
      <button onClick={() => setLaugh({ a: "ㅅㅈ" })}>지우기</button>
      <hr />
      <h1>{laugh}</h1>
    </div>
  );
};

export default Laugh;
```

App.js

```
import React from "react";
import Laugh from "./laugh";

const App = () => {
  return <Laugh />;
};

export default App;
```

1.3 Effect Hook

1.3.1 useEffect(렌더링 될 때 마다)

kwonTA.js

```
import React, { useEffect, useState } from "react";

const KwonTA = () => {
  const [css, changeCss] = useState({
    fontSize: "10pt",
    width: "300px",
    height: "300px",
    resize: "None",
  });

  useEffect(() => {
    alert("야");
  });

  return (
    <div>
      <textarea
        style={css}
        onChange={(e) => {
          changeCss({ ...css, fontSize: e.target.value + "pt" });
        }}
      />
    </div>
  );
};

export default KwonTA;
```

App.js

```
import React from "react";
import KwonTA from "./kwonTA";

const App = () => {
  return <KwonTA />;
};

export default App;
```

1.3.2 useEffect(첫 렌더링때만)

kwonTA.js

```
import React, { useEffect, useState } from "react";

const KwonTA = () => {
  const [css, changeCss] = useState({
    fontSize: "10pt",
    width: "300px",
    height: "300px",
    resize: "None",
  });

  useEffect(() => {
    alert("야");
  }, []);

  return (
    <div>
      <textarea
        style={css}
        onChange={(e) => {
          changeCss({ ...css, fontSize: e.target.value + "pt" });
        }}
      />
    </div>
  );
};

export default KwonTA;
```

1.3.3 useEffect(특정 값 바뀔때만)

kwonTA.js

```
import React, { useEffect, useState } from "react";

const KwonTA = () => {
  const [css, changeCss] = useState({
    fontSize: "10pt",
    width: "300px",
    height: "300px",
    resize: "None",
  });
  const [css2, changeCss2] = useState({ fontSize: "10pt" });

  useEffect(() => {
    alert("야");
  }, [css2]);

  return (
    <div>
      <textarea
        style={css}
        onChange={(e) => {
          changeCss({ ...css, fontSize: e.target.value + "pt" });
        }}
      />
      <hr/>
      <button
        style={css2}
        onClick={(e) => {
          changeCss2({ fontSize: css2.fontSize === "10pt" ? "30pt" : "10pt" });
        }}
      >
        하이 빅스비
      </button>
    </div>
  );
};

export default KwonTA;
```

1.3.4 useEffect(cleanup - 시작/끝)

kwonTA.js

```
import React, { useEffect, useState } from "react";

const KwonTA = () => {
  const [css, changeCss] = useState({…
  });
  const [css2, changeCss2] = useState({ fontSize: "10pt" });

  useEffect(() => {
    alert("끝 : " + css2.fontSize);
    return () => {
      alert("시작 : " + css2.fontSize);
    };
  }, [css2]);

  return (
    <div>
      <textarea
        style={css}…
        }}
      />
      <hr />
      <button
        style={css2}
        onClick={(e) => {
          changeCss2({ fontSize: css2.fontSize === "10pt" ? "30pt" : "10pt" });
        }}
      >
        하이 빅스비
      </button>
    </div>
  );
};

export default KwonTA;
```

1.4 useRef

kwonCV.js

```
import React, { useRef } from "react";

const KwonCV = () => {
  const paper = useRef(null);
  const moveMove = (e) => {
    const pen = paper.current.getContext("2d");
    pen.fillStyle = "white";
    pen.fillRect(0, 0, 500, 500);
    pen.fillStyle = "black";
    pen.fillRect(e.clientX - 25, e.clientY - 25, 50, 50);
  };
  return (
    <div>
      <canvas
        ref={paper}
        onMouseMove={moveMove}
        width={"500px"}
        height={"500px"}
      ></canvas>
    </div>
  );
};

export default KwonCV;
```

App.js

```
import React from "react";
import KwonCV from "./kwonCV";

const App = () => {
  return <KwonCV />;
};

export default App;
```

2. 커스텀 Hook

2.1 커스텀 Hook

useKwonProduct.js

```
import { useReducer } from "react";

function updateProduct2(s, p) {
  return {
    ...s,
    [p.name]: p.value,
  };
}
//ednf
export default function useKwonProduct() {
  const [product, updateProduct] = useReducer(updateProduct2, {
    name: "",
    price: "",
  });
  const setProduct = (e) => {
    updateProduct(e.target);
  };
  return [product, setProduct];
}
```

useKwonProducts.js

```
import { useReducer } from "react";

function addProduct2(products, p) {
  return products.concat(p);
}

export default function useKwonProducts() {
  const [products, addProduct] = useReducer(addProduct2, []);

  const chugaProduct = (p) => {
    addProduct({ name: p.name, price: p.price });
    p.name = "";
    p.price = "";
  };
  return [products, chugaProduct];
}
```

2.2 커스텀 Hook 활용

snackBBS.js

```
import React from "react";
import useKwonProduct from "./useKwonProduct";
import useKwonProducts from "./useKwonProducts";

const SnackBBS = () => {
  const [snacks, addSnack] = useKwonProducts();
  const snackList = snacks.map((s, i) => (
    <tr key={i}>
      <td>{s.name}</td> <td>{s.price}</td>
    </tr>
  ));
  const [snack, setSnack] = useKwonProduct();

  return (
    <>
      <table border={1}>
        <tr>
          <th>이름</th> <th>가격</th>
        </tr>
        {snackList}
      </table>
      <hr />
      <input name="name" value={snack.name} onChange={setSnack} />
       
      <input name="price" value={snack.price} onChange={setSnack} />
       
      <button
        onClick={() => {
          addSnack(snack);
        }}
      >
        추가
      </button>
    </>
  );
};

export default SnackBBS;
```

App.js

```
import React from "react";
import SnackBBS from "./snackBBS";

const App = () => {
  return <SnackBBS />;
};

export default App;
```